AI-Powered Stock Market Insight Generator

INDIVIDUAL THESIS

Fernando K C S

IT21471162

Individual component: Financial Document Understanding and Language-Driven Insight Extraction

Group ID: R26-IT-076

Report submitted in partial fulfilment of the requirements for the

BSc (Hons) in Information Technology

Information Technology

Department of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

October

2026

DECLARATION AND SUPERVISOR CERTIFICATION

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Student name: Fernando K C S

Student ID: IT21471162

Signature: Chanodya Date: 09/10/2026

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Supervisor: Dr. Kapila Dissanayaka

Signature: _____ Date: _____

Co-supervisor: Dr. Mahima Weerasinghe

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ABSTRACT

Annual reports contain information that is important to investors but difficult to interpret because it is distributed across financial statements, notes, governance disclosures, risk sections, and narrative commentary. Component 2 develops a source-grounded financial document understanding pipeline that extracts report text and tables page by page, identifies investor-relevant information, constructs expected outputs with page-level evidence, and benchmarks ten prompt-engineering strategies using a shared JSON schema. The system evaluates field coverage, numeric accuracy, semantic similarity, source faithfulness, investor relevance, and hallucination penalty. Across six evaluated reports, the recorded benchmark selected the Beginner Investor Explanation Prompt as the best overall strategy with an average final score of 0.1693, coverage of 0.9342, investor relevance of 1.0000, and a source-faithfulness score of 0.0000. The results demonstrate that prompt design can improve structured coverage and user-oriented explanations, while also exposing the need for stronger evidence verification and numeric normalization. The component produces non-advisory investor insights and is integrated with a dashboard that supports annual and interim report analysis.

Keywords: annual report analysis, PDF extraction, prompt engineering, source-grounded insight, investor education

ACKNOWLEDGEMENTS

I acknowledge the guidance of my supervisor and the support of the project team during the design, implementation, testing, and documentation of Component 2. I also acknowledge the contribution of open-source document-processing and natural-language-processing libraries used in the implementation. All generated outputs were reviewed as software artefacts, and the limitations of automated extraction and language-model processing are reported in this thesis.

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

API - Application Programming Interface

CSE - Colombo Stock Exchange

JSON - JavaScript Object Notation

LLM - Large Language Model

OCR - Optical Character Recognition

PDF - Portable Document Format

1 INTRODUCTION

1.1 Overall project context

The wider project is an investor-oriented financial intelligence platform combining market forecasting, financial report understanding, news and sentiment analysis, and external-market risk analysis. Component 2 provides the document-understanding layer that converts lengthy company disclosures into structured facts and concise explanations.

1.2 Individual component context

Component 2 focuses on annual and interim company reports. It verifies report metadata, extracts page-level text and tables, retrieves evidence relevant to investor questions, and returns structured information covering company overview, performance, financial position, cash flow, ratios, shareholder information, risks, governance, and outlook.

1.3 Problem statement

Retail investors face a high information burden because important facts are separated across many pages and expressed in inconsistent formats. Generic summarisation may omit numbers, merge unrelated disclosures, or produce statements without verifiable evidence. This component addresses how to provide accessible explanations while retaining a traceable connection to the source report.

1.4 Research gap

Existing work addresses financial question answering, layout-aware document understanding, risk extraction, and KPI extraction. A practical gap remains in localized, source-grounded extraction for company reports in an emerging-market setting. This component addresses the gap through controlled prompt comparison and measurable source-grounded evaluation.

1.5 Aim and objectives

The aim is to develop and evaluate a source-grounded pipeline for extracting investor-relevant information from company reports. Objectives are to extract report content

with page references, discover relevant evidence, construct expected outputs, benchmark ten prompts, calculate evaluation metrics, select a best prompt, and generate clear non-advisory insights.

1.6 Research questions or hypotheses

RQ1: Can page-level extraction and retrieval provide reusable evidence for investor-related report analysis?

RQ2: Do prompt strategies produce different coverage, numeric accuracy, similarity, faithfulness, and relevance scores?

RQ3: Can the selected prompt generate understandable explanations without becoming investment advice?

1.7 Scope limitations and assumptions

The scope is limited to text-based reports available as PDFs. OCR is not enabled by default, so scanned documents may require preprocessing. Complex tables may not be represented perfectly. The system assumes that page text can be extracted and that a report contains enough context to identify its company and reporting period.

1.8 Individual contribution

The individual contribution is the design and implementation of PDF extraction, metadata detection, retrieval, expected-output construction, prompt definitions, benchmark execution, metric calculation, best-prompt selection, final insight generation, dashboard integration, and automated tests.

1.9 Thesis structure

Chapter 2 reviews related work. Chapter 3 presents methodology. Chapter 4 describes requirements and architecture. Chapter 5 explains implementation and testing. Chapter 6 reports results. Chapter 7 discusses implications and limitations. Chapter 8 addresses sustainability, and Chapter 9 concludes the thesis.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review covers financial question answering, document layout understanding, information extraction from annual reports, risk disclosure extraction, and prompt-engineering evaluation.

2.2 Domain and theoretical foundations

Financial report understanding combines document processing, information retrieval, structured extraction, and natural-language generation. Source grounding is treated as a reliability requirement: a generated fact should be supported by an extracted passage and page number.

2.3 Related research

Financial question-answering work such as FinQA demonstrates numerical reasoning over financial statements. Layout-aware models such as DocLLM show the value of document structure. Other research investigates risk and KPI extraction. Component 2 extends these directions with prompt comparison and localized investor-facing explanations.

2.4 Existing systems products and techniques

Keyword retrieval is transparent but can miss paraphrases. Embedding retrieval improves semantic matching but may select less interpretable context. Pure summarisation is easy to use but may omit evidence. The selected design combines deterministic extraction and retrieval with structured model output.

2.5 Gap synthesis and conceptual framework

The framework is report PDF to page-level extraction, extraction to investor-information discovery, discovery to evidence-backed expected output, expected output to ten prompt variants, prompt outputs to evaluation, and evaluation to best-prompt selection and final insight generation.

3 RESEARCH METHODOLOGY

3.1 Research design

The study uses an artefact-based experiment. The same extracted report context and output schema are provided to ten prompt strategies. Prompt strategy is the independent variable; coverage, numeric accuracy, semantic similarity, source faithfulness, investor relevance, and hallucination penalty are dependent variables.

3.2 Data participants or study materials

The materials are company annual reports stored as PDFs. The recorded benchmark evaluated six reports, including DIAL 2024, JKH 2022, JKH 2023, JKH 2024, LOLC 2024, and an additional report identified as 369_1741169817422. No human participant data is required for the core benchmark.

3.3 Experimental or study procedure

The extractor records page text, tables, headings, and warnings. Retrieval identifies likely pages, the expected-output builder records values and evidence, ten prompts are executed, outputs are normalized and scored, and the highest average final score is selected for final insight generation.

3.4 Measures and evaluation metrics

Field coverage measures expected fields extracted. Numeric accuracy measures agreement after unit and currency normalization. Semantic similarity compares expected and generated values. Source faithfulness checks cited quotes against extracted text. Investor relevance measures focus on useful categories, while unsupported claims attract a hallucination penalty.

3.5 Data analysis

The final score is 0.25 coverage + 0.20 numeric accuracy + 0.20 semantic similarity + 0.20 source faithfulness + 0.15 investor relevance - hallucination penalty. Scores are clamped between zero and one, and prompt rankings use the average score across reports.

3.6 Ethics privacy security and responsible AI

The system is informational and non-advisory. Credentials are supplied through environment variables. The schema preserves missing values rather than forcing unsupported claims. Users are told to check insights against the original report and that outputs are not financial advice.

3.7 Reproducibility and research artefacts

The component stores extracted pages, expected outputs, prompt scores, benchmark tables, the selected prompt, and final insights separately. Tests cover extraction, schemas, metadata, unsupported source quotes, API-key handling, and insight fusion.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The system accepts report PDFs, detects identity and period, extracts text and tables, retrieves evidence, produces a stable JSON schema, compares prompts, selects a best prompt, generates readable insights, and exports artefacts. Traceability, reproducibility, privacy, and safe handling of missing information are non-functional requirements.

4.2 Component architecture within the complete solution

The internal pipeline contains PDF extraction, metadata detection, retrieval, expected-output construction, prompt benchmarking, evaluation, best-prompt selection, and final insight generation. The dashboard presents report facts and insights alongside market, news, and risk components.

4.3 Design alternatives and decisions

A pure keyword system was rejected because it is weak for paraphrases. Free-form responses were rejected because they are difficult to compare and validate. The selected design uses deterministic extraction, a fixed schema, ten controlled prompts, explicit missing fields, and separate evaluation artefacts.

4.4 Detailed design

The schema separates extracted facts from investor-friendly insight. Fields may be null when reliable information is unavailable. Source evidence is stored separately so factual values and explanatory prose do not become conflated.

4.5 Standards and professional practice

The design follows separation of concerns, environment-based configuration, schema validation, automated testing, explicit limitations, and evidence-based reporting. It does not present language-model output as verified financial advice.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The component is implemented in Python with modules for PDF extraction, retrieval, normalization, prompt execution, evaluation, and insight generation. Streamlit provides the interactive layer, while JSON and Markdown store reproducible artefacts.

5.2 Component implementation

The extractor creates page-level JSON. Metadata detection identifies company, report type, and period. The expected-output builder records source snippets. Ten prompt strategies cover basic extraction, role framing, schema constraints, evidence, verification, ratios, risk, beginner explanations, few-shot prompting, and hybrid best practice.

5.3 Integration with the other components

The integrated application invokes Component 2 through report validation and report-insight services. The report insight is combined with market forecasts, news and sentiment, and external-market risk. The component also runs independently for research and testing.

5.4 Verification and software testing

Automated tests cover extraction, expected-output schema, prompt-output schema, metadata, local analysis, insight fusion, unsupported source quotes, API-key privacy,

and the uploaded-report adapter. These tests target malformed JSON, fabricated citations, credential leakage, and integration failures.

5.5 Deployment and operation

Reports are placed in the annual_reports directory, followed by extraction, expected-output construction, benchmarking, evaluation, prompt selection, and final insight generation. In the integrated runtime, uploaded reports are validated and a deterministic summary remains available if the optional language service is unavailable.

6 RESULTS AND VALIDATION

6.1 Research results

The ten-prompt experiment evaluated six reports. Prompt 08, the Beginner Investor Explanation Prompt, achieved the best average final score of 0.1693, with coverage 0.9342 and investor relevance 1.0000. Numeric accuracy was 0.0619, semantic similarity 0.1168, source faithfulness 0.0000, and hallucination penalty 0.25. The results show strong coverage but insufficient evidence verification.

6.2 Product and technical results

The component produces page-level extraction files, expected outputs, prompt scores, a comparison table, the selected prompt, and final insight reports. The stable schema supports both dashboard rendering and future evaluation.

6.3 User or stakeholder validation

The current artefacts provide technical validation rather than a completed human-user study. The selected prompt was designed for beginner investors and recorded investor relevance of 1.0000. A future study should measure comprehension, trust, task time, and evidence retrieval.

6.4 Integration results

The dashboard exposes report analysis alongside the other project components. Report validation occurs before insight generation, and deterministic summaries support graceful degradation when a model service is unavailable.

6.5 Summary of findings

Prompt strategy affects coverage and explanation style. High coverage alone is not sufficient for a trustworthy financial document system; evidence validation, numeric normalization, and citation checking require further work.

7 DISCUSSION

7.1 Interpretation of findings

The beginner-oriented prompt ranked first because it maintained strong coverage and investor relevance. Its low final score shows why evaluation must include faithfulness and numeric checks rather than relying on readability or completeness alone.

7.2 Comparison with literature baselines and existing solutions

Compared with generic summarisation, the component provides a fixed schema, missing-field handling, page-level extraction, and prompt comparison. Compared with financial QA, it covers broader report categories. Compared with layout-aware systems, its table handling is simpler and remains a limitation.

7.3 Research contribution

The research contribution is a reproducible framework for comparing prompts for localized investor-related annual report extraction, including a shared schema, evidence-backed expected outputs, weighted evaluation, and reusable prompt selection.

7.4 Product contribution

The product contribution is an operational report-analysis component that transforms uploaded reports into structured facts and understandable non-advisory insights, with fallback behaviour and automated tests.

7.5 Limitations and threats to validity

The benchmark is limited by available reports and heuristic expected outputs. The source-faithfulness score of zero requires investigation, and low numeric accuracy suggests that financial units, signs, and table representations need stronger normalization. Results should not be generalized to every report format or model.

7.6 Implications

Developers should implement schema and evidence checks before optimizing explanation style. Investors may use the system to locate information but should still read the source report. Future research needs stronger retrieval, table parsing, and human-validated reference outputs.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The value proposition is faster and more accessible navigation of lengthy disclosures. Target users include beginner investors, students, educators, analysts, and organizations needing a report overview without direct trading recommendations.

8.2 Market and competitive position

The component combines localized report analysis, a measurable prompt benchmark, page-level evidence artefacts, and non-advisory explanations. General document assistants and specialist financial-data platforms are competitive alternatives; transparency is the main differentiator.

8.3 Business or sustainability model

A sustainable deployment could offer report analysis as a subscription or institutional service with document-size limits. Educational deployments can use deterministic extraction and cached outputs. Reusable evaluation artefacts reduce maintenance when prompts or models change.

8.4 Scalability intellectual property and compliance

Scalability requires asynchronous processing, caching, document-size controls, OCR, and stronger table services. Compliance requires disclaimers, secure credential handling, audit logs, and correction procedures for unsupported insights.

8.5 Contribution to the Sustainable Development Goals

The component can support SDG 4 through financial and digital literacy, SDG 8 through access to business information, and SDG 9 through applied document-intelligence infrastructure, provided that outputs remain accessible and non-advisory.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

Component 2 developed a source-grounded financial document understanding pipeline combining PDF processing, retrieval, expected-output construction, prompt benchmarking, evaluation, prompt selection, and insight generation.

9.2 Achievement of objectives

The component achieved the objectives of extracting report content, identifying investor fields, constructing structured outputs, benchmarking ten prompts, calculating metrics, selecting a best prompt, and integrating report insights. It also exposed weaknesses in source faithfulness and numeric accuracy.

9.3 Individual research and product contributions

The research contribution is the measured comparison of ten prompt strategies. The product contribution is the tested Component 2 implementation, dashboard integration, stored artefacts, deterministic fallback, and non-advisory insight format.

9.4 Future directions

Future work should add OCR, layout-aware table extraction, stronger citation validation, improved currency and unit normalization, human-validated gold outputs, multilingual support, and a user study.

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GLOSSARY

Source-grounded insight: an explanation whose material claims are linked to evidence extracted from the source report.

Prompt strategy: a defined instruction pattern used to guide the language model for the same report context.

Investor relevance: the degree to which an output addresses information useful for understanding a company and its disclosures.

Hallucination penalty: a score reduction applied to unsupported, unverifiable, or source-inconsistent claims.

APPENDICES

Appendix A Research instruments or extended results

The repository contains the full prompt-score CSV and JSON files, extracted page payloads, expected outputs, and final insight artefacts. These files should be included with the submission or referenced through the project repository as required by the supervisor.

Appendix B Academic integrity and AI use disclosure

Tools used include Python document-processing libraries, PDF extraction libraries, the project language-model adapter, and automated test tooling. AI-assisted outputs were treated as draft material and checked against source artefacts and implementation behaviour. The author remains responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this thesis.